Coherence Is Not Accuracy: Evaluating Reconciliation, Calibration, and Reliability-Aware View Selection in Tabular Foundation Models

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ABSTRACT

Tabular foundation models can return incompatible probabilities when two categorical outcomes are queried in different orders. A projected joint distribution can satisfy their marginal constraints exactly, but this does not establish that it predicts unseen outcomes better. We study this distinction with CoRe-TFM, which places raw chain-rule factorizations, pooling, marginal reconciliation and validation-based selection in a common evaluation. Controlled studies isolate when preserving the predicted marginals helps and when it transfers their errors into the joint forecast. A separate fixed-test benchmark evaluates TabICLv2 and TabPFN-3 on ten paired-target datasets, with 160 model–dataset–split–context configurations. At the 256-example training context, hard projection toward the directly predicted marginals reduced structural discrepancy in all 80 model–dataset–split comparisons, yet increased test negative log-likelihood (NLL) in 67. The dataset-balanced mean NLL increases were 0.01831 for TabICLv2 and 0.01178 for TabPFN-3. Changing the projection reference from arithmetic to geometric pooling produced a similar descriptive pattern. A simple validation-selected raw-view-or-pool policy achieved slightly lower mean NLL than fixed arithmetic pooling, but the difference was small and not established by multiplicity-adjusted exploratory tests. A guarded repair policy never activated. Six selected model configurations reproduced their archived predictions exactly in a fresh GPU execution. Official-source checksum reports cover all eleven OpenML records, while exact historical preprocessing and original loaded-weight identity remain incompletely authenticated. A consistency constraint therefore needs predictive validation: eliminating an algebraic contradiction does not, by itself, improve a forecast.

1. Introduction

Suppose a tabular foundation model is asked to predict two categorical outcomes, A and B, for the same record. It can predict each outcome directly, or it can predict one outcome after conditioning on a possible value of the other. Probability theory requires the answers to be compatible with a single joint distribution. A model answering the queries separately has no such guarantee.

A mismatch between the views poses two different problems. We can ask whether the probabilities are algebraically compatible, or whether a compatible replacement would score better on future observations. The first can be decided from the predicted distributions. The second cannot: it requires outcome data. A post-processing rule may resolve the contradiction and still assign less probability to the events that occur.

This issue is relevant to tabular in-context learning. Prior-Data Fitted Networks and the TabPFN family use pretrained models for tabular prediction (Müller et al., 2022; Hollmann et al., 2023, 2025; Grinsztajn et al., 2025, 2026); TabICLv2 offers a separate open approach (Qu et al., 2026). Benchmarks such as TabArena compare their predictive performance (Erickson et al., 2025), while work on uncertainty estimation and proper scoring rules shows why classification accuracy alone is insufficient to assess their probabilities (Landsgesell and Knoll, 2026; Johnson et al., 2026; Costa et al., 2026; Tanna et al., 2026).

For categorical A and B and tabular features X, consider four queried views:

$$p(A \mid X), \qquad p(B \mid X), \qquad p(A \mid B, X), \qquad p(B \mid A, X). \tag{1}$$

They yield two chain-rule joint forecasts:

$$J^{B\to A}(a, b \mid x) = p_B(b \mid x)p_{A|B}(a \mid b, x), \tag{2}$$

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$$J^{A\to B}(a, b \mid x) = p_A(a \mid x)p_{B|A}(b \mid a, x). \tag{3}$$

If all four predictions came from one joint distribution, the two expressions would agree, and marginalizing either expression would recover the appropriate direct prediction. In practice, neither property is guaranteed. These discrepancies have recently been studied directly for tabular foundation models (Klötergens et al., 2026). The broader problem of recovering a joint distribution from possibly incompatible conditionals is well established (Majid et al., 2025; Muré, 2019).

One response is to combine the two joint forecasts, then project the combination onto the direct marginals. Such a projection enforces the chosen constraints; it does not establish their truth. Indeed, proper scoring rules distinguish a coherent but inaccurate forecast from an accurate one (Brier, 1950; Gneiting and Raftery, 2007; Gneiting et al., 2007). Forecast-combination research makes a similar distinction between agreement among inputs and the quality of the resulting predictive distribution (Genest and Zidek, 1986; Ranjan and Gneiting, 2010; Gneiting and Ranjan, 2013).

We use CoRe-TFM to compare unmodified chain-rule forecasts with pooled and reconciled distributions, and to ask whether validation labels can guide that choice. Synthetic experiments provide ground truth, making the reliability assumption behind projection testable. Two real-model designs supply complementary evidence: a five-seed robustness study and a matched fixed-test study in which the training context changes without changing the evaluation examples. Both report proper scores; neither treats improved consistency as a substitute for held-out prediction.

Contributions.

1. We give a precise distinction between compatibility and predictive risk, including a strict-propriety counterexample and controlled perturbations of marginal and conditional reliability.

2. We compare established marginal-projection and pooling operations with raw-view selection and calibration on tabular foundation-model predictions. The contribution is the evaluation and reliability question, not a claim to have invented iterative proportional fitting.

3. We report a five-seed robustness study and a separate 160-configuration, ten-dataset fixed-test evaluation, making differences in model versions, candidate families and validation budgets explicit.

4. We measure the score changes caused by direct repair, examine cases where repair helps, assess the failure of a conservative repair gate to activate, and document a six-configuration clean GPU replication with explicit provenance qualifications.

2. Related Work

2.1. Tabular foundation models and probability reliability

Prior-Data Fitted Networks established a pretrained approach to tabular inference (Müller et al., 2022). TabPFN and its subsequent versions, and independently developed TabICLv2, have expanded this approach (Hollmann et al., 2023, 2025; Grinsztajn et al., 2025, 2026; Qu et al., 2026). Their relative advantages depend on the dataset and evaluation regime; tree ensembles remain important baselines (Gorishniy et al., 2021; Grinsztajn et al., 2022). TabArena supplies a continuously maintained benchmark context (Erickson et al., 2025).

Much of the TFM literature compares models using aggregate predictive scores. That is useful but does not resolve how the probabilities from overlapping queries should be combined. ScoringBench examines proper scores for probabilistic TFM regression and illustrates how model rankings depend on the evaluation criterion (Landsgesell and Knoll, 2026). Regression-focused uncertainty studies raise a related concern about the reliability of predictive distributions (Johnson et al., 2026; Costa et al., 2026); they are not direct evaluations of the paired categorical outcome considered here. An analysis of TFM ensembles also reports that improvements in classification accuracy may come at a cost in log loss (Tanna et al., 2026). Closest to our problem, Klötergens et al. (2026) examine failures of marginalization and factorization consistency. Here we evaluate the consequences of repairing those failures using held-out predictive scores.

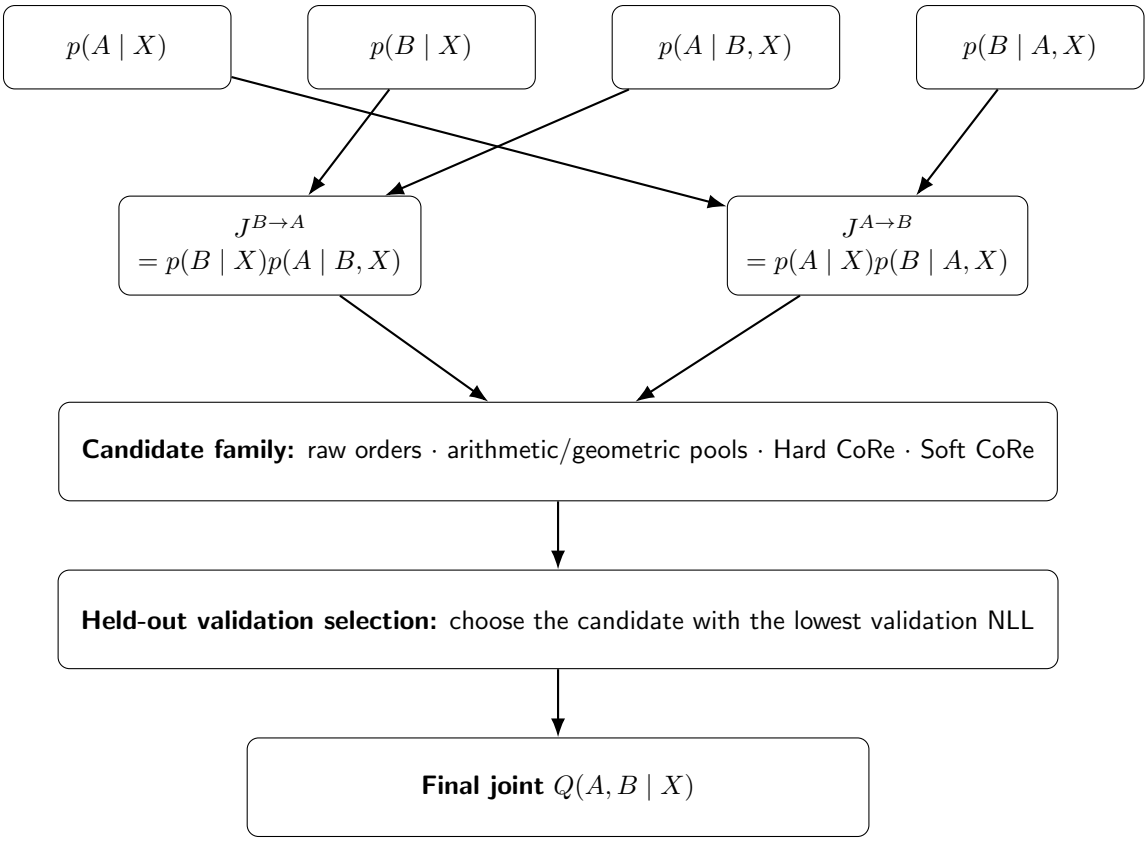


Figure 1: CoRe-TFM workflow. Marginal and conditional queries yield two chain-rule joint distributions. Raw, pooled and reconciled forecasts are evaluated separately; where selection is used, its decision is based on held-out validation labels rather than test outcomes.

2.2. Pooling, calibration and knowledge fusion

Linear and logarithmic opinion pools are classical ways to combine distributions (Genest and Zidek, 1986; Heskes, 1997). A pool is not guaranteed to preserve the calibration of its components, and transformed pooling can sometimes improve forecasts (Ranjan and Gneiting, 2010; Gneiting and Ranjan, 2013). Accordingly, we evaluate arithmetic and geometric consensus predictions against held-out outcomes rather than treating pooling as a neutral preprocessing step.

The logarithmic and Brier scores are strictly proper rules for evaluating predictive probabilities (Brier, 1950; Gneiting and Raftery, 2007). Calibration is also distinct from sharpness (Gneiting et al., 2007). Temperature, beta and Dirichlet calibration offer different post-hoc ways of adjusting probability estimates using validation labels (Guo et al., 2017; Kull et al., 2017, 2019). Evidence from calibrated random forests illustrates how strongly calibration gains can depend on the underlying estimator and data (Shaker and Hüllermeier, 2025).

Knowledge merging supplies a related perspective on conflicting information (Hunter and Summerton, 2006); supervised dynamic fusion is another way of learning how to weight imperfect sources (Zhang and Lin, 2025). Our four probability views, however, are not independent measurements or experts. They are overlapping queries made to the same model, so agreement among them should not be interpreted as independent corroboration.

2.3. Conditional compatibility and marginal projection

Conditionals specified separately need not correspond to one joint distribution. Both compromises between incompatible conditionals and formal conditions for consistent recovery have been studied (Muré, 2019; Majid et al., 2025). In our setting, the two chain-rule reconstructions make part of this conflict measurable. Their agreement does not reveal which reconstruction is closer to the unknown data-generating distribution.

Hard CoRe uses an information projection with prescribed marginals. The numerical machinery is well established: iterative proportional fitting, I-divergence geometry, matrix scaling, and related Bregman and optimal-transport scaling algorithms (Deming and Stephan, 1940; Csiszár, 1975; Sinkhorn, 1964; Cuturi, 2013; Benamou et al., 2015; Chizat et al., 2018). The question studied here is not whether projection can satisfy the constraints, but whether imposing those particular model-predicted marginals improves a forecast. This also creates a selection problem: with many candidate post-processors and few validation labels, a method can overfit the selection criterion (Cawley and Talbot, 2010). We therefore compare methods on paired held-out outcomes and aggregate repeated experiments at dataset level (Demšar, 2006).

3. Problem Formulation

Let $A \in \{1, \dots, K_A\}$ and $B \in \{1, \dots, K_B\}$. For an input x, a frozen predictor supplies the four views described above. We define the factorization inconsistency

$$\Delta_{\mathrm{fac}}(x) = \mathrm{TV}\big(J^{B\to A}, J^{A\to B}\big) = \frac{1}{2}\sum_{a,b}\Big|J^{B\to A}(a,b \mid x) - J^{A\to B}(a,b \mid x)\Big|. \tag{4}$$

The two induced joints preserve different direct marginals automatically: $J^{B\to A}$ preserves p_B and $J^{A\to B}$ preserves p_A. Their opposite marginals may disagree with the direct queries.

Our objective is a normalized joint $Q(A, B \mid X)$ that balances three different notions:

1. *compatibility*: structural agreement among the queried views;
2. *fidelity*: closeness of the post-processed joint to the model's queried distributions; and
3. *predictive quality*: proper-score performance against held-out outcomes.

These notions need not agree. In particular, zero post-repair inconsistency is not evidence that Q is more accurate.

A strict-propriety counterexample. Let R be a positive pooled joint distribution and $Q = \mathcal{P}_{p_A,p_B}(R) \neq R$. Under logarithmic loss, if the true joint distribution is $P^* = R$, the expected score change due to repair is

$$\mathbb{E}_{Y\sim R}[-\log Q(Y)] - \mathbb{E}_{Y\sim R}[-\log R(Y)] = \mathrm{KL}(R\|Q) > 0. \tag{5}$$

Conversely, if $P^* = Q$, the same repair decreases expected log loss by $\mathrm{KL}(Q\|R) > 0$. This is a direct consequence of strict propriety, not a new optimization theorem. It demonstrates why marginal compatibility cannot by itself fix the sign of the predictive effect: the additional assumption needed is that the imposed marginals, and hence the resulting joint, are reliable relative to the unmodified forecast.

The four views are repeated predictions from one frozen model and therefore overlap statistically. They must not be interpreted as independent Bayesian experts. CoRe objectives are regularized compatibility objectives, not likelihood multiplication under conditional independence.

4. CoRe-TFM

4.1. Pooling baselines

For a direction weight $w \in [0, 1]$, the arithmetic pool is

$$Q_{\mathrm{arith},w} = wJ^{B\to A} + (1-w)J^{A\to B}. \tag{6}$$

It minimizes $w\,\mathrm{KL}(J^{B\to A}\|Q) + (1-w)\,\mathrm{KL}(J^{A\to B}\|Q)$. The normalized geometric pool is

$$M_w(a,b) = \frac{J^{B\to A}(a,b)^w J^{A\to B}(a,b)^{1-w}}{\sum_{a',b'} J^{B\to A}(a',b')^w J^{A\to B}(a',b')^{1-w}}, \tag{7}$$

which is the corresponding reverse-KL barycenter.

Table 1
Relation to adjacent methods. "Outcome-gated" means that held-out labels may determine whether a transformation is applied. These categories summarize typical uses and are not claims that every published variant has identical properties.

Approach	Multiple probability views	Enforces joint compatibility	Uses held-out labels	Outcome-gated	Primary target
Calibration methods	No	No	Yes	No	Calibration / proper score
Opinion pooling	Yes	No	Optional	No	Consensus forecast
TFM self-consistency diagnostics	Yes	No	No	No	Detect inconsistency
Hard marginal projection	Yes	Yes	No	No	Structural coherence
CoRe-TFM	Yes	Optional	Yes	Yes	Reliability-aware predictive quality

4.2. Hard CoRe

Define the transportation polytope with prescribed direct marginals

$$\mathcal{T}(p_A, p_B) = \left\{ Q \geq 0 : \sum_b Q(a,b) = p_A(a),\ \sum_a Q(a,b) = p_B(b) \right\}. \tag{8}$$

Let R be a strictly positive reference joint, with the same class support as the four views. The hard marginal reconciliation operator is

$$\mathcal{P}_{p_A,p_B}(R) = \arg\min_{Q \in \mathcal{T}(p_A,p_B)} \mathrm{KL}(Q \| R). \tag{9}$$

In Protocol A the reference is the normalized geometric pool $R = M_w$. Protocol B uses the equal-weight arithmetic pool $R = Q_{\mathrm{arith},0.5}$. Since the projection depends on its reference distribution, these interventions are distinguished throughout the evaluation. The feasible set is nonempty because $p_A p_B$ satisfies the marginal constraints. With strictly positive R, the projection is unique and can be calculated by iterative proportional fitting. What is assumed, but not guaranteed by the optimization, is that the prescribed direct marginals are reliable. Unless stated otherwise, "hard repair" in the fixed-test results means $\mathcal{P}_{p_A,p_B}(Q_{\mathrm{arith},0.5})$.

4.3. Soft CoRe

To relax that assumption, Soft CoRe solves

$$\begin{aligned} Q_{\mathrm{soft}} = \arg\min_Q\ & w\,\mathrm{KL}(Q \| J^{B\to A}) + (1-w)\,\mathrm{KL}(Q \| J^{A\to B}) \\ & + \lambda_A\,\mathrm{KL}(Q_A \| p_A) + \lambda_B\,\mathrm{KL}(Q_B \| p_B). \end{aligned} \tag{10}$$

The chain rule for KL divergence yields

$$\mathrm{KL}(Q \| J^{B\to A}) = \mathrm{KL}(Q_B \| p_B) + \mathbb{E}_{b\sim Q_B}\,\mathrm{KL}(Q_{A|B=b} \| p_{A|B=b}), \tag{11}$$

$$\mathrm{KL}(Q \| J^{A\to B}) = \mathrm{KL}(Q_A \| p_A) + \mathbb{E}_{a\sim Q_A}\,\mathrm{KL}(Q_{B|A=a} \| p_{B|A=a}), \tag{12}$$

showing that all four predictive views enter the objective. Generalized scaling uses exponents

$$\tau_A = \frac{\lambda_A}{1+\lambda_A}, \qquad \tau_B = \frac{\lambda_B}{1+\lambda_B}. \tag{13}$$

As $\lambda_A, \lambda_B \to \infty$, the method approaches hard marginal preservation; at small penalties it stays closer to the factorized consensus.

4.4. Selective CoRe

Selective CoRe treats reconciliation as optional. The candidate family includes the two raw orders, equal and weighted arithmetic/geometric pools, weighted Hard CoRe, and weighted Soft CoRe. The implemented grid uses five direction weights

$$\{0, 0.25, 0.5, 0.75, 1\} \tag{14}$$

and seven marginal penalties

$$\{0.03, 0.1, 0.3, 1, 3, 10, 30\}. \tag{15}$$

After duplicate removal, the full family contains 48 candidates. The candidate with the lowest held-out validation NLL is frozen before test evaluation.

For a fixed finite candidate family $\mathcal{C}$ of size M, let all candidates be specified independently of the validation labels, with per-example negative log-likelihood bounded by $B = -\log \epsilon$ after a positive probability floor ϵ. If the validation observations are i.i.d. and L is the corresponding population loss, a union-bound/Hoeffding argument gives

$$L(\hat{Q}) \leq \min_{Q \in \mathcal{C}} L(Q) + 2B\sqrt{\frac{\log(2M/\delta)}{2n_v}}, \tag{16}$$

with probability at least $1 - \delta$. This is a conservative finite-class bound for the selection stage, not a guarantee about the upstream foundation model, non-independent examples, adaptively constructed candidates, or out-of-distribution deployment. The dependence on both M and the validation size n_v motivates the candidate-family and validation-fraction analyses.

4.5. Opportunity and selection regret

To separate method headroom from selection failure, let L_{arith} be arithmetic-pool loss, L_{sel} the selected loss, and L_{oracle} the minimum test loss over the candidate family. We report

$$\text{oracle opportunity} = L_{\text{arith}} - L_{\text{oracle}}, \tag{17}$$

$$\text{selection regret} = L_{\text{sel}} - L_{\text{oracle}}, \tag{18}$$

$$\text{net effect} = L_{\text{sel}} - L_{\text{arith}}. \tag{19}$$

The oracle is analysis-only and is never used to choose a deployable candidate.

5. Experimental Design

5.1. Controlled known-truth studies

We use three controlled layers. First, a multiclass surrogate data-generating process provides exact $P^*(A, B \mid X)$ and therefore exact expected proper scores. Second, an exact-view perturbation benchmark independently corrupts the two direct marginals and the two conditional families, isolating the reliability assumptions encoded by reconciliation. Third, a heterogeneous 100-task mixture varies view reliability from task to task and compares Selective CoRe with fixed methods and a full-family oracle. A separate validation-size study varies the number of labels available to the selector.

5.2. Real-model benchmark

The real-data analysis includes Anneal, Car, Credit, Customer, Diamonds, Marketing, MIC, Nursery, Phishing, and Wine. The Car and Wine benchmarks follow the UCI and wine-quality sources, respectively (Bohanec, 1988; Cortez et al., 2009). Two released TFMs form the primary model comparison: TabICLv2 2.1.1 and TabPFN 8.4.0 using the TabPFN-3 interface. CatBoost 1.2.10 is an explicitly non-TFM boundary baseline and is excluded from estimates labeled "primary TFM."

TabFM was excluded from the frozen experimental protocol for recorded runtime and operational compatibility reasons, not for unfavorable predictive performance. The release required by the adapter was not fully exercised in this bounded execution environment, and no controlled TabFM timing comparison exists in the evidence archive. It would therefore be inappropriate to infer a slowdown factor or relative accuracy from its absence.

The evaluation uses five outer folds per dataset and model, disjoint training, validation and test partitions, training-set coverage of the target classes, and no more than 128 test observations in a fold. At a 256-row training cap there are 52 validation observations per fold. For one seed the three-model matrix has 10 datasets $\times$ 3 models $\times$ 5 folds $=$ 150 model–fold tasks and 1,200 reported method rows (eight retained method summaries per task). The 48 candidate distributions inside Selective CoRe should not be confused with those eight reported methods.

5.3. Seed and data-budget robustness

The robustness dataset contains five complete runs at the 256-row cap, using constrained-sampling seeds 11, 23, 42, 71 and 101. Four additional caps (64, 128, 512 and 1024) were evaluated with each of three matched seeds (23, 42 and 71), for a total of 17 runs. The archived aggregate contains 20,400 method-result rows: 17 runs × 10 datasets × 3 models × 5 folds × 8 methods. We used a fixed validation fraction of approximately 20%, giving 13, 26, 52 and 103 validation observations at caps of 64, 128, 256 and 512, respectively. At a nominal cap of 1024, available training data limited the actual cap to 718–1024 observations and validation support to 144–205 (mean 194.4).

The budget experiment is deliberately described as *combined upstream-and-selection evidence sensitivity*: changes to training context and validation support cannot be causally disentangled. The 256-row row uses five seeds unless explicitly marked as the matched-three-seed subset; other caps use three.

5.4. Fixed-test, matched-context evaluation

Protocol B separately varies training context and selector labels, which cannot be disentangled in the five-seed budget study. It also includes calibrated-pooling baselines. Because the split construction, model package versions and candidate families differ, Protocol B is analyzed independently of Protocol A. The fixed-test archive comprises ten dataset pairs (the same named benchmark collection), two released TFMs (TabICLv2 2.2.0 and TabPFN package 9.2.0 using `ModelVersion.V3`), four randomized held-out split settings per dataset (seed/fold 23/0, 23/1, 42/0, 71/0), and two nested training contexts of 128 and 256 examples. This yields $10 \times 2 \times 4 \times 2 = 160$ real-model prediction caches. The split setting named "fold" is a randomized repeat, not a classical disjoint K-fold partition. The two protocols also differ in their treatment of some feature–target pairs, most notably the Diamonds `cut/color` construction. Raw-source checksum verification does not establish the identity of all historical feature transformations, so effects from the two protocols are not pooled.

For each dataset and split setting, 205 held-out validation examples and 128 held-out test examples are reserved independently of the context cap. Both TFMs and both context sizes use identical validation and test identifiers within the setting; nested context subsets protect the paired comparison. The 205-label validation pool alongside 128 or 256 context examples deliberately *decouples* upstream training context from selection evidence; it is not a 128-total-label learning scenario. This protocol has at least 205 validation labels and 128 test labels in addition to its stated training context. It therefore probes contextual inference and selector sample size rather than ordinary limited-label learning with 128 or 256 total observed labels. We report the protocol-designated central comparison at the 256-example cap and the first 52 validation labels, and use the 128-example cap as a context sensitivity analysis. Per-example NLL, multiclass joint Brier and classification accuracy are scored on the unchanged test set; top-label ECE is descriptive.

Four probabilistic queries are inferred using separately fitted marginal and conditional estimators. The independent audit reconstructs each raw chain-rule joint and the arithmetic/geometric pools, hard marginal projection, partial soft repair, and validation-fitted selectors. Protocol B uses eight base candidates rather than Protocol A's 48-candidate research formulation. Consequently, a "full-family" result in Protocol B means selection among those eight candidates. The three partial repairs are convex interpolations, $Q_\alpha = (1-\alpha)Q_{\text{arith},0.5} + \alpha \mathcal{P}_{p_A,p_B}(Q_{\text{arith},0.5})$, with $\alpha \in \{0.25, 0.50, 0.75\}$. They are *not* numerical solutions of the KL-penalized Soft CoRe objective above. The two formulations are reported under their respective names rather than treated as interchangeable. The validation-only baselines include raw-or-pool selection, temperature-fitted pooling, and a guarded repair option. Temperature and policy choice use no test labels. For a given split, all candidate losses are scored on the same fixed held-out examples.

For the principal inferential contrasts, we first average four repeated split effects within each dataset separately for each model, then report the unweighted mean over ten datasets. We give paired dataset-resampling bootstrap intervals and exploratory two-sided Wilcoxon tests, followed by Holm adjustment over the documented primary exploratory policy family. The 160 caches and their 1,920 reconstructed method rows are not interchangeable statistical sample sizes; only ten datasets are treated as independent experimental units for dataset-blocked comparisons. Protocol B records package versions, checkpoint identifiers, frozen split hashes and cache SHA-256 values. The later replication establishes file fingerprints for its own execution, but not the loaded-weight identity of every original benchmark job.

5.5. Targeted replication and dataset provenance

After freezing the 160-cache analysis, we performed a targeted reproduction in a fresh Colab GPU runtime using the archived train, validation and test IDs. For each of Car, Nursery and Diamonds, both TabICLv2 and TabPFN-3 were run at training context 256, seed 23 and split index 0, yielding six additional executions of existing configurations. No additional dataset or split was added to the primary $n = 160$ benchmark. The replication compared saved probability

Table 2
Two real-model experimental protocols retained as separate evidence streams. The second protocol isolates changes in training context at fixed validation/test IDs; its four randomized split settings are not independent datasets.

Component	Protocol A: five-seed benchmark	Protocol B: fixed-test benchmark	Interpretive consequence
Models	TabICLv2 2.1.1; TabPFN 8.4.0/V3 interface	TabICLv2 2.2.0; TabPFN 9.2.0/V3	Do not pool model-version estimates
Datasets	Ten paired-target datasets	Ten paired-target datasets	Paired identities require feature/preprocessing review
Sampling	Five outer folds; five seeds at 256	Four randomized held-out settings	Different replication units and split construction
Contexts	64–1024 in budget study	128 and 256 with nested context	Fixed test IDs across contexts in Protocol B
Selection labels	Proportional to context (52 at 256)	205 reserved; primary first 52	Selection and context budgets vary separately in Protocol B
Selector family	Up to 48 candidates	Eight base candidates plus calibration policies	Different method-family sizes
Statistical unit	Dataset (ten)	Dataset (ten)	Multiple runs do not create independent datasets

tensors and proper scores against the historical caches, recorded the software environment, and fingerprinted the explicitly configured model-file paths. These file hashes document the replication execution; historical cache metadata did not contain authenticated loaded-weight hashes. In a separate online step, we requested the metadata and raw source files for the eleven underlying OpenML records (the Wine pairing joins two source IDs) and compared downloaded MD5 values against the OpenML-reported values. We retain the result of each request, including the initial failed MIC request, rather than silently substituting a different dataset.

5.6. Outcomes, aggregation and inference

Joint negative log-likelihood (NLL) and joint Brier score are the principal proper scores. Additional diagnostics include conditional NLL, expected calibration error (ECE), total variation between the two factorizations, and distortion of the direct marginals. ECE depends on the selected binning and is descriptive here; known-truth experiments also permit exact expected loss, truth distance and oracle regret.

For a method comparison, we first form the paired score difference within each dataset–model–fold–seed cell. Differences are averaged over folds, over the two TFMs, and then over seeds within each dataset. The appropriate unit for the reported paired inference is the *dataset* ($n = 10$), not the individual fold, row, seed, or model cell. A two-sided Wilcoxon signed-rank test is applied to the ten dataset-level paired effects. Because the study contains exploratory choices among methods, budgets and data regimes, the reported p-values should not be interpreted as independently preregistered confirmatory tests. We report model-specific and budget comparisons as exploratory; no family-wise confirmatory inference is claimed for the many additional contrasts (Demšar, 2006). The five-seed 256-row bootstrap interval reported below resamples ten datasets with replacement (20,000 percentile replicates; fixed analysis seed 20261012); it is a sensitivity interval, not a pre-registered equivalence test. Positive "Selective-minus-arithmetic" NLL differences favor arithmetic pooling.

Display precision and practical scale. Displayed losses and differences are rounded for readability. The source CSV/JSON files in S1 retain the unrounded values used in the real-model analyses. An NLL difference also has a direct multiplicative interpretation. If $\Delta = L_{\text{method}} - L_{\text{baseline}}$, then $\exp(-\Delta)$ is the ratio of the geometric-mean probability assigned to the observed class by the method and the baseline. Thus, the controlled Selective-CoRe improvement of about 0.0067 NLL corresponds to roughly a 0.67% increase in geometric-mean probability on the observed class, whereas the five-seed real-data disadvantage of about 0.0009 corresponds to only about a 0.09% decrease. The larger 0.1505-NLL failure of compulsory Hard CoRe corresponds to about a 14.0% decrease. These are multiplicative

probability interpretations, not absolute percentage-point changes, and downstream practical value remains application- and decision-cost-dependent.

6. Results

6.1. Exact-view perturbations expose the reliability assumption

Table 3 summarizes the frozen, known-truth perturbation analysis reported in the original controlled-study artifacts. These controlled results are distinguished from the 17-run real-model archive audit; a fresh end-to-end synthetic rerun is not claimed here.

When the direct marginals are reliable, Hard CoRe improves over arithmetic pooling by 0.0133 NLL on average ($p_{\text{Holm}} = 7.11 \times 10^{-15}$). When the conditionals are reliable but direct marginals are noisy, Hard CoRe is worse by 0.1505 ($p_{\text{Holm}} = 7.11 \times 10^{-15}$). Exact marginal preservation therefore transfers the quality of the marginals into the joint; it is not intrinsically beneficial.

6.2. Heterogeneous reliability can reward adaptive selection

Table 4 presents the original heterogeneous 100-task controlled benchmark. Its task-level evidence is analyzed separately from the ten-dataset, dataset-blocked real-model comparison.

Selective CoRe improves over arithmetic pooling by 0.0067 mean exact NLL, wins 68/100 tasks, and has a paired Wilcoxon $p = 1.25 \times 10^{-11}$. It selects Soft CoRe 38 times, geometric pooling 33 times, raw orders 19 times, arithmetic pooling 10 times, and Hard CoRe zero times. Its mean regret to the non-deployable full-family oracle is only 0.0007. The controlled validation-size study is consistent with the finite-selection interpretation: mean oracle regret falls from 0.0056 with 100 validation labels to 0.0012 with 800 labels.

6.3. Frozen real benchmark: coherence improves while proper scores worsen

The original frozen 256-row run produced the result in Table 5. This is retained because it was the first prespecified real-model analysis and because the later robustness study changes its interpretation rather than invalidating its execution.

Selective CoRe improves agreement with the direct marginals while worsening both proper scores in this frozen run. The result demonstrates the main conceptual point: a structurally more faithful joint can be predictively worse. Mean factorization TV in the primary TFM cells is nonzero, so the finding is not caused by a trivial absence of incompatibility.

The effect is strongly model dependent. In the frozen run, Selective CoRe is worse than arithmetic by 0.0067 NLL on TabICLv2 but better by 0.0030 on TabPFN-3. The paired difference between model-specific effects is 0.0097 ($p = 0.00977$, exploratory).

6.4. Five-seed robustness attenuates the frozen negative result

The completed multi-seed analysis repeats the 256-row protocol for five constrained-sampling seeds. After averaging folds, the two primary TFMs and the five seeds within each dataset, the mean Selective-minus-arithmetic effect is +0.000851 NLL. Selective CoRe is better on two datasets and worse on eight. The two-sided dataset-blocked Wilcoxon result is $p = 0.10547$. An exploratory percentile bootstrap over the ten dataset means gives a 95% interval of approximately [−0.00157, +0.00291] (20,000 replicates, fixed seed 20261012). This interval includes both small benefits and small harms and should not be read as establishing equivalence. The five seed-level mean effects are +0.0004, +0.0009, +0.0018, +0.0009, and +0.0002 for seeds 11, 23, 42, 71, and 101, respectively.

This robustness result does not establish equivalence between Selective CoRe and arithmetic pooling. It does show that the magnitude and inferential strength of the original single-seed disadvantage were sensitive to the sampling realization. The defensible 256-row conclusion is therefore that Selective CoRe has a small average disadvantage under this bounded protocol, with substantial data-set and model heterogeneity and without dataset-blocked significance over the five-seed aggregate.

6.5. Data-budget sensitivity reveals a sign transition

Table 6 summarizes the primary TFM effect over the available data budgets. For the 64, 128, 512, and 1024 caps, three matched seeds are available; the 256 row contains five seeds. A matched-three-seed calculation gives 0.0012 rather than 0.0009 at 256 and leaves the qualitative sequence unchanged.

These are descriptive comparisons, not five separate confirmatory claims of improvement. The experiment contains ten dataset units, some datasets cannot supply the nominal 1024 training rows, and each larger context also supplies more

Table 3
Best mean exact expected NLL in each exact-view perturbation regime (50 paired seeds).

Regime	Best method	Expected NLL
Uniform low noise	Geometric pool	1.4786
Direct marginals reliable	Soft CoRe ($\lambda = 10$)	1.4868
Conditionals reliable	Arithmetic pool	1.5088
$B \to A$ reliable	$J^{B \to A}$	1.4753
$A \to B$ reliable	$J^{A \to B}$	1.4753
All four views noisy	Arithmetic pool	1.5387

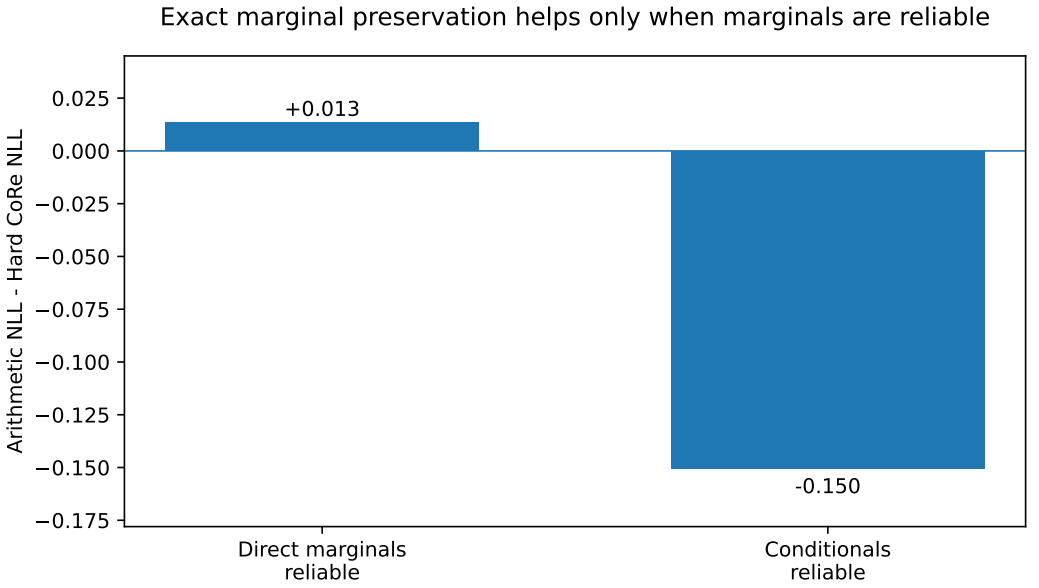


Figure 2: Reliability mechanism in the exact-view perturbation study. The vertical axis is arithmetic NLL minus Hard-CoRe NLL, so positive values favor exact marginal preservation. Hard CoRe helps when direct marginals are reliable and fails sharply when the conditionals are reliable but the direct marginals are noisy.

held-out validation examples. The negative difference at the largest cap is therefore evidence of a budget-dependent association in this benchmark, not a causal demonstration that either training context or validation size alone makes reconciliation profitable.

The model-specific pattern strengthens the reliability interpretation (Table 7). TabPFN-3 is near parity at 64 rows and favors Selective CoRe at every larger budget in the aggregate. TabICLv2 begins with a substantially larger selector penalty and crosses to a small selector advantage only at 512 rows.

6.6. Inconsistency is not a universal reconciliation-gain signal

Factorization TV decreases as the evidence budget grows for both TFMs. For TabICLv2, mean TV falls from 0.1237 at the 64-row cap to 0.0560 at 1024; for TabPFN-3 it falls from 0.0601 to 0.0403. Direct and conditional NLLs also improve with increasing budget.

Crucially, the relationship between inconsistency and Selective-CoRe benefit is not stable across models. In a frozen benchmark reliability analysis, the Pearson correlation between factorization TV and arithmetic-minus-selective gain is approximately -0.421 for TabICLv2 and $+0.310$ for TabPFN-3, with an overall correlation of only about -0.133 across all model cells. A high factorization TV therefore does not, by itself, tell us whether reconciliation will improve proper-score performance. The diagnostic must be interpreted together with model- and view-specific reliability.

Table 4
Heterogeneous 100-task controlled benchmark.

Method	Mean NLL	SE	Mean oracle regret
Selective CoRe	**1.4837**	0.0055	**0.0007**
Arithmetic	1.4903	0.0056	0.0074
Geometric	1.4914	0.0056	0.0084
Soft CoRe ($\lambda = 1$)	1.5006	0.0060	0.0176
$J^{A\to B}$	1.5068	0.0070	0.0239
$J^{B\to A}$	1.5144	0.0067	0.0315
Hard CoRe	1.5383	0.0085	0.0554

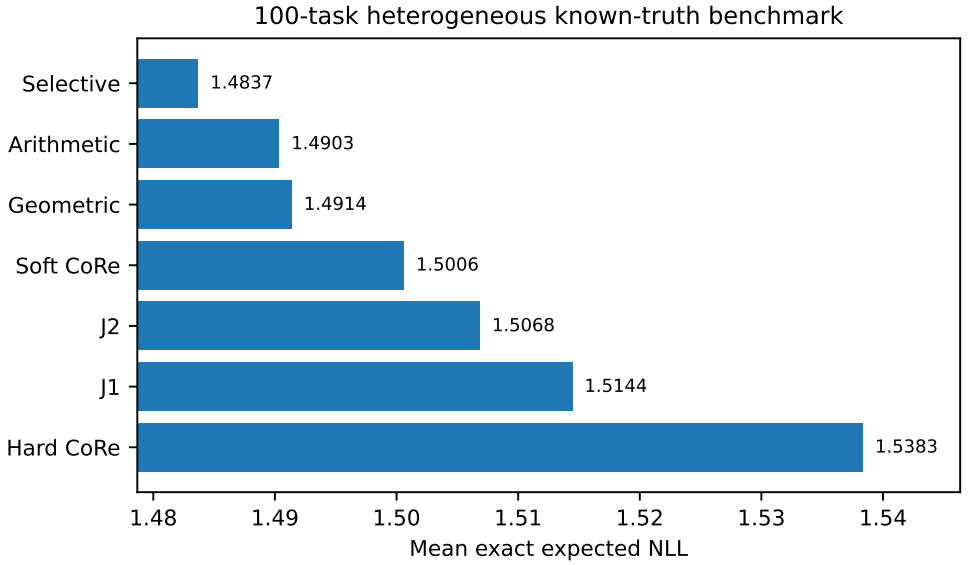


Figure 3: Mean exact expected NLL in the 100-task heterogeneous known-truth benchmark. Selective CoRe has the lowest mean NLL and mean oracle regret, while compulsory Hard CoRe is the worst fixed strategy in this deliberately heterogeneous reliability mixture.

Table 5
Frozen single-seed dataset-blocked Selective-CoRe effects relative to arithmetic pooling across the two primary TFMs ($n = 10$ datasets). Positive proper-score differences are worse.

Metric	Mean difference	95% bootstrap CI	Wins	p
Joint NLL	+0.0018	[+0.0005, +0.0033]	2/10	0.02734
Joint Brier	+0.0006	—	1/10	0.00586
Marginal distortion	−0.0039	—	9/10	0.00391

This conclusion is also supported analytically: one can construct pairs of factorizations with large TV where one factorization is already equal to the truth and pooling necessarily hurts, as well as pairs with arbitrarily small disagreement symmetrically bracketing the truth where pooling helps. Inconsistency is a structural defect statistic, not an accuracy statistic.

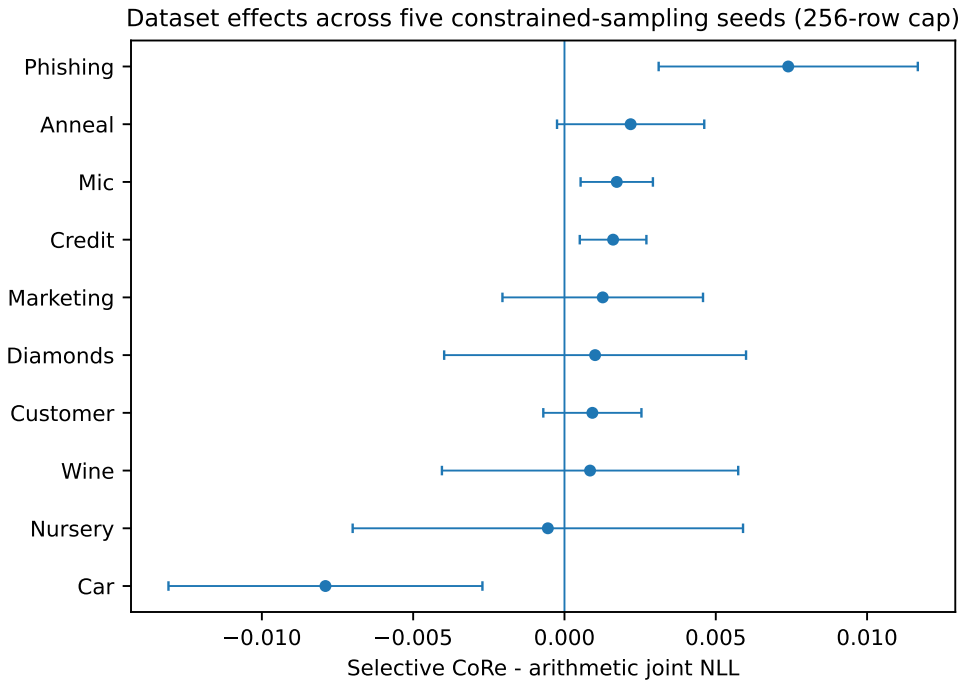


Figure 4: Dataset-level Selective-minus-arithmetic joint-NLL effects at the 256-row cap across five constrained-sampling seeds. Points show the seed mean and horizontal bars show one standard deviation across seeds. Negative values favor Selective CoRe.

Table 6
Selective-minus-arithmetic joint-NLL effect as the training cap and proportional validation pool increase. Positive values favor arithmetic pooling. The 256-row aggregate uses five seeds; other rows use three.

Train cap	Typical validation	Mean effect	TFM cells	Direction
64	13	+0.0065	60	Arithmetic better
128	26	+0.0025	60	Arithmetic better
256	52	+0.0009	100	Near parity
512	103	−0.0013	60	Selective better
1024	≈ 194	−0.0034	60	Selective better

6.7. Candidate-family complexity is measurable

The full 48-candidate selector is not uniformly superior to a smaller family. Table 8 reports mean joint NLL for nested candidate-family ablations aggregated over the robustness runs.

Pool-only selection is slightly better than the full family from 64 through 512 rows, while the full family becomes marginally best at 1024. This is consistent with a bias–variance trade-off in candidate selection: a larger family creates more oracle opportunity but also more ways to overfit a finite validation set.

At the 256-row cap, a separate nested-validation sensitivity gives mean NLL 1.3805, 1.3798, and 1.3786 when using 25%, 50%, and 100% of the available validation pool, respectively. At the 64-row cap, using the full validation pool also improves mean NLL relative to the two smaller fractions. These ablations support the role of validation evidence while avoiding the stronger claim that every increase in validation size must improve every dataset.

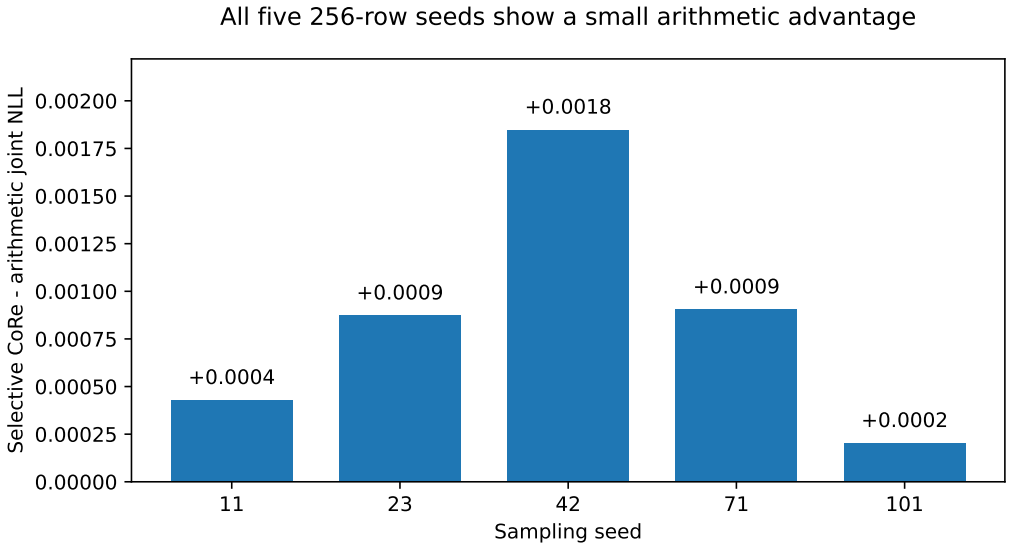


Figure 5: Seed-level Selective-minus-arithmetic NLL effects at the 256-row training cap. All five sampling seeds yield a small positive mean effect, but the magnitude varies materially across seeds.

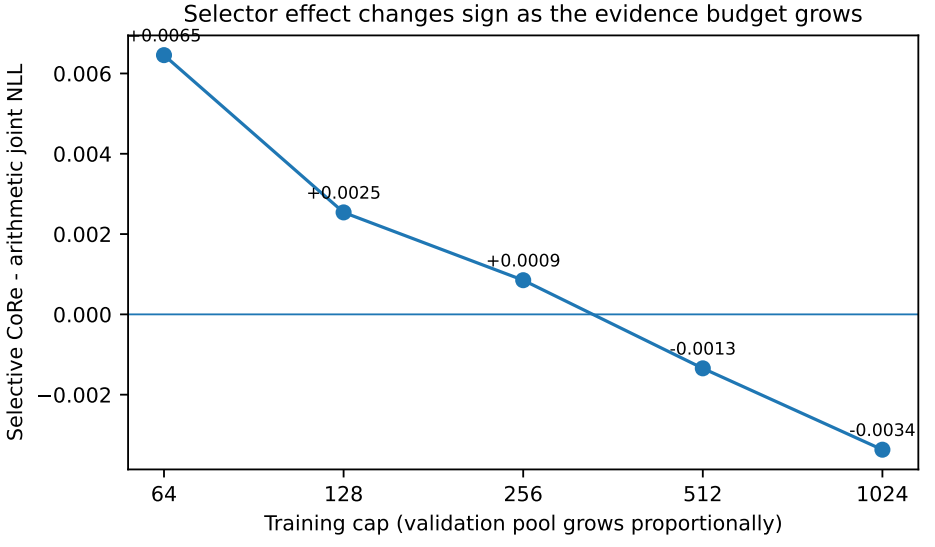


Figure 6: Evidence-budget transition. Positive values favor arithmetic pooling; negative values favor Selective CoRe. Training context and the held-out validation pool grow together, so this is an evidence-budget trend rather than a causal context-size effect.

6.8. Rare classes do not explain the 256-row effect

Some paired-target datasets contain very small target-class supports. We therefore perform a post-hoc exclusion sensitivity using data-set-level minimum target-class support. At the 256-row cap, keeping all ten datasets gives mean Selective-minus-arithmetic NLL +0.0009. Requiring minimum support of at least 2 leaves nine datasets and yields +0.0008; requiring at least 5 leaves six datasets and yields +0.0010; requiring at least 10 leaves five datasets and yields +0.0007. None of the filtered analyses reverses the direction. Rare-class datasets therefore do not appear to be the sole explanation for the small 256-row disadvantage.

Table 7
Model-specific mean Selective-minus-arithmetic NLL difference.

Model	64	128	256	512	1024
TabICLv2	+0.0120	+0.0056	+0.0044	−0.0004	−0.0027
TabPFN-3	+0.0009	−0.0005	−0.0026	−0.0023	−0.0041

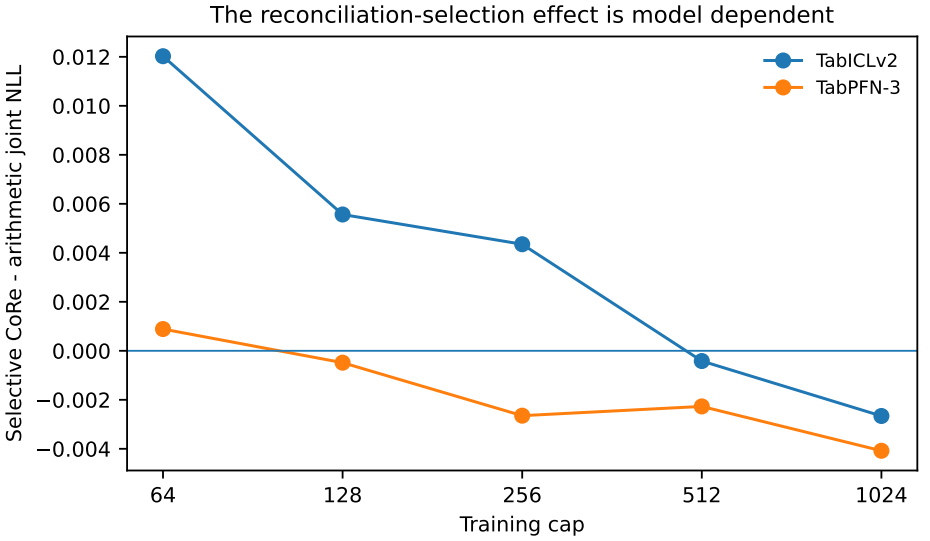


Figure 7: Model-specific evidence-budget effects. TabPFN-3 reaches a Selective-CoRe advantage earlier, while TabICLv2 pays a much larger small-budget selection cost and crosses the zero-effect line only at larger budgets.

Table 8
Mean joint NLL of validation-selected candidate families. Lower is better.

Train cap	Full	Pool only	Raw only	Repair only
64	1.6884	**1.6867**	1.6906	1.6885
128	1.4913	**1.4911**	1.4932	1.4912
256	1.3786	**1.3779**	1.3802	1.3788
512	1.2854	**1.2853**	1.2868	1.2854
1024	**1.2089**	1.2093	1.2103	1.2089

6.9. A conservative Safe-Selective rule is a negative control, not a win

The repository includes a Safe Selective CoRe rule based on structural-risk penalties. It falls back to arithmetic pooling unless a validation improvement exceeds the candidate-family complexity penalty. In 100 controlled known-truth tasks, the current rule selects the arithmetic fallback in all 100 cases, has zero mean difference from arithmetic, and mean oracle regret 0.0065. This establishes a useful negative control: a theoretically motivated guardrail can be so conservative that it eliminates both selection risk and selection benefit. We therefore do not present the current Safe-Selective rule as a performance contribution.

6.10. Ten-dataset fixed-test evaluation

We audited all 160 real-model cache/metadata pairs from the second protocol, checked 40 frozen split specifications, and recomputed each method's test scores directly from saved probability arrays. The primary comparison uses training

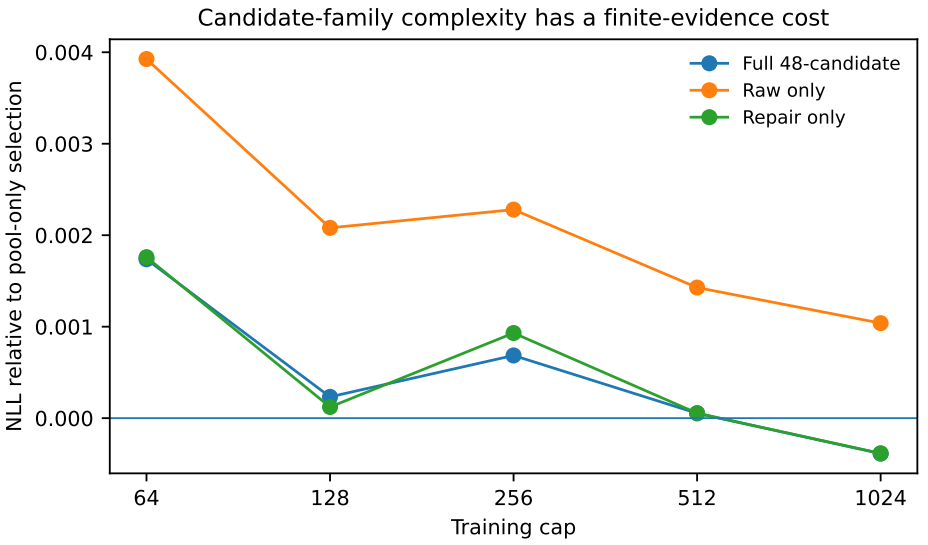


Figure 8: Candidate-family selection cost relative to the pool-only selector. Values above zero are worse than pool-only selection. The full 48-candidate family pays a measurable finite-evidence penalty at small and moderate budgets and becomes slightly better only at the largest budget.

cap 256, validation budget 52, four matched split settings per dataset, and ten dataset-blocked means per TFM. Table 9 gives mean joint NLL, with lower values indicating better proper-score performance.

The raw-or-pool selector shows the smallest mean NLL in both models, but the margin relative to fixed arithmetic pooling is only −0.000813 in TabICLv2 and −0.002212 in TabPFN-3. Among the 40 primary configurations per model, the restricted full selector chose hard or partial repair in 4 TabICLv2 cases and 8 TabPFN-3 cases. These are *selection events*, not evidence of repair benefit; averaged performance still does not beat the smaller raw-or-pool family. Dataset-blocked exploratory Wilcoxon p-values for these differences are 0.6953 and 0.2324, respectively, and their bootstrap intervals include zero. Hence the data do not establish reliable predictive superiority of the selector. Nor does the Protocol B full-family selection produce a demonstrable gain over the smaller raw-or-pool family. Notably, temperature fitting on a 52-example validation subset *worsens mean test NLL* under the current grid and splits; this is a finding about the implemented finite-label calibration procedure, not a general argument against calibration. The accompanying proper-score and decision metrics are displayed in Table 10; their aggregation preserves dataset balance and the distinction between log loss, squared-probability error, top-1 decisions and calibration diagnostics.

6.11. Matched training-context sensitivity under fixed validation and test IDs

The second protocol holds each split's 205-example validation pool and 128-example test pool fixed while increasing the model's nested training context from 128 to 256 rows. This isolates a different factor from the earlier experiment in which upstream context and validation evidence changed together. Table 11 reports within-split paired changes in test NLL, with negative differences indicating improvement from the larger context. For both models, arithmetic pooling improves on 39 of 40 paired split configurations and on all ten dataset means. The corresponding mean decreases are 0.13332 (TabICLv2) and 0.11163 (TabPFN-3) NLL units. This reflects the benefit of additional task context under the recorded setup; it does not imply that a reconciliation policy is necessary or that the total label budget is limited to 256 examples.

6.12. Direct structural intervention: coherence is not accuracy

For each fixed test prediction, we apply the arithmetic-reference marginal projection from Eq. (9), holding the queried direct marginal distributions fixed. The marginal-discrepancy statistic is $\frac{1}{2}[\mathrm{TV}(Q_A, p_A) + \mathrm{TV}(Q_B, p_B)]$ averaged over test cases; it measures agreement *with the model's own direct predictions*, not distance to the unknown true marginal distribution. We independently measure whether this intervention decreases mean marginal total variation and whether the resulting held-out NLL improves. All 80 primary model–dataset–split configurations show reduced marginal

Table 9
Fixed-test benchmark: unweighted mean over ten datasets of joint NLL at training cap 256 and validation budget 52 (four paired split settings per dataset). *Full-family* denotes the fixed-test restricted eight-candidate selector, not the older 48-candidate method. Bold denotes the lowest descriptive mean in each model column, not a significant win.

Policy	TabICLv2	TabPFN-3
Arithmetic pool	1.34000	1.34147
Raw-or-pool selector	**1.33919**	**1.33926**
Full-family selector	1.33994	1.33944
Temperature-calibrated pool	1.35350	1.35492
Hard reconciliation	1.35831	1.35325
Halfway soft repair	1.34654	1.34524
Guarded calibrated repair	1.35350	1.35492

Table 10
Secondary proper-score and decision metrics in the new fixed-test protocol at training context 256, averaged equally across ten datasets (four split repeats within each dataset). NLL and multiclass Brier are lower-is-better, top-1 accuracy is higher-is-better, and ten-bin top-label ECE is an unstable descriptive diagnostic. Values are not significance tests.

TFM	Method	NLL ↓	Brier ↓	Acc. ↑	ECE ↓
TabICLv2	Arithmetic	1.3400	0.6148	0.4869	0.0669
TabICLv2	Raw-or-pool	1.3392	0.6144	0.4893	0.0671
TabICLv2	Restricted full selector	1.3399	0.6147	0.4877	0.0678
TabICLv2	Hard repair	1.3583	0.6180	0.4828	0.0679
TabICLv2	Halfway partial repair	1.3465	0.6157	0.4848	0.0693
TabICLv2	Temperature scaling	1.3535	0.6185	0.4869	0.0807
TabPFN-3	Arithmetic	1.3415	0.6142	0.4846	0.0555
TabPFN-3	Raw-or-pool	1.3393	0.6139	0.4824	0.0590
TabPFN-3	Restricted full selector	1.3394	0.6140	0.4828	0.0574
TabPFN-3	Hard repair	1.3533	0.6162	0.4848	0.0530
TabPFN-3	Halfway partial repair	1.3452	0.6147	0.4824	0.0544
TabPFN-3	Temperature scaling	1.3549	0.6176	0.4846	0.0712

Table 11
Fixed-test, matched-context NLL differences, 256 minus 128 context rows. Each model has 40 matched split configurations (ten datasets, four repeats), but the reported descriptive paired test uses ten dataset-averaged differences. These results do not isolate model pretraining or architecture effects.

Model	Policy	Mean ΔNLL	Improved splits	Improved datasets
TabICLv2	Arithmetic pool	-0.13332	39/40	10/10
TabICLv2	Raw-or-pool	-0.13404	39/40	10/10
TabPFN-3	Arithmetic pool	-0.11163	39/40	10/10
TabPFN-3	Raw-or-pool	-0.10986	39/40	10/10

discrepancy, but 67/80 show *increased* NLL. The reduction is expected from the construction of a marginal projection and should not be counted as an independent external-validation success; the held-out score comparison is the substantive empirical test. This isolates a substantive distinction: enforcing the selected marginal constraints changes the joint as intended, but the constraints may not be the most reliable information available about the true joint.

The corresponding mean Brier increases are +0.00324 for TabICLv2 and +0.00199 for TabPFN-3. These bootstrap intervals exclude zero and the unadjusted paired p-values are below 0.05, yet both Holm-adjusted p-values exceed 0.05. Consequently we treat the widespread hard-repair penalty as a strong descriptive pattern with exploratory statistical support, rather than a family-wise-confirmed positive finding. The Holm values are exploratory per-model adjustments over the specified candidate/reference contrasts; they do not exhaust all historical tests, budgets, models or additional

Table 12
Audited paired hard-reconciliation effects relative to arithmetic pooling. Positive ΔNLL/ΔBrier means worse predictive scoring. Bootstrap intervals and exploratory tests use ten dataset-level average effects; Holm correction covers the documented exploratory contrast family.

Model	Mean ΔNLL	Bootstrap 95% CI	Harmed	Wilcoxon p	Holm p
TabICLv2	+0.01831	[+0.00893, +0.02832]	9/10	0.00977	0.06836
TabPFN-3	+0.01178	[+0.00397, +0.01954]	9/10	0.01953	0.13672

post-hoc questions examined during research development. Bootstrap intervals excluding zero are likewise not evidence of universal protection against exploratory multiplicity. Partial (50%) soft repair reduces but does not eliminate the mean harm: its dataset-blocked ΔNLL is +0.00654 for TabICLv2 and +0.00377 for TabPFN-3. The intervention demonstrates a mechanism, but does not establish that all possible coherence-enforcing maps have the same effect.

6.13. Sensitivity to the projection reference distribution

The Protocol A hard projection and the Protocol B direct ablation use different KL-reference distributions: the Protocol A equal-weight variant projects the normalized geometric pool, whereas the fixed-test audit projects the arithmetic pool. To test whether the observed failure mode is an artifact of that choice, we performed an additional *post-hoc, inference-free* ablation using the frozen predictions from all 160 real-model configurations. The geometric-reference projection uses $\mathcal{P}_{p_A,p_B}(M_{0.5})$ with the same iterative proportional-fitting tolerances as the arithmetic-reference intervention. No checkpoint was reloaded, no hyperparameter was selected using test labels, and both projected joints were scored on the same saved held-out outcomes.

The geometric-reference projection also increases dataset-average NLL on nine of ten datasets for each model. Its arithmetic-pool-relative mean changes are +0.01800 for TabICLv2 and +0.01126 for TabPFN-3; both projections drive marginal discrepancy close to zero. The two reference distributions give numerically different repaired predictions, but their dataset-level patterns agree in this setting. The mean geometric-minus-arithmetic-reference repaired NLL changes are −0.00031 and −0.00052 respectively. These post-hoc results strengthen the *reference-sensitivity check* without making the historical penalty-based Soft CoRe or its 48-candidate selector equivalent to the restricted October analysis. No multiplicity-adjusted confirmatory claim is made for these new contrasts.

6.14. Dataset heterogeneity and the Diamonds exception

The sign of the repair effect is not universal. Across the ten datasets, hard projection harms mean NLL on nine datasets per model but improves it on Diamonds (mean ΔNLL −0.00513 for TabICLv2 and −0.01028 for TabPFN-3). Split-level analysis shows that TabICLv2's Diamonds improvement is driven by a large gain in the 71/0 split (−0.04562), while its other three splits are harmed. TabPFN-3 improves in three of the four Diamonds splits. Class-pair subgroup sizes can be sparse, so the observed advantage is an informative heterogeneous case, *not* proof of a broadly identifiable beneficial-repair regime. No dataset-specific rule was learned from test outcomes.

6.15. Guarded repair and evidence-budget findings

The validation-guarded repair policy did not activate in its recorded runs and exactly matched its temperature-calibrated-pool fallback. It is therefore inappropriate to call the guard *effective at identifying beneficial repairs*; its observed behavior demonstrates nonactivation and an inability to outperform the fallback, not a safety guarantee under distribution shift. Likewise, the Protocol A 48-candidate selection results and the fixed-test eight-base-candidate results refer to different candidate sets and environment versions. Both experiments caution against treating a larger candidate family as intrinsically better when validation evidence is limited.

The two evidence-budget designs provide complementary, noninterchangeable evidence. In the historical protocol, training context and selection support increase together and the Selective-minus-arithmetic trend changes sign by the largest cap. In the audited fixed-test protocol, 128 versus 256 training contexts are nested within an unchanged 205-example validation pool and unchanged 128-example test pool, allowing context-size sensitivity to be studied independently of label availability for policy selection. Neither design constitutes a prospective deployment or independent-hospital evaluation, and neither is a confirmation that calibration or selection dominates all other post-processing strategies.

Table 13
Exploratory reference-distribution ablation at training context 256. Differences are dataset-balanced test NLL changes; positive values indicate harm. The last column counts datasets with worse geometric-reference repair than arithmetic pooling. The geometric reference corresponds to the equal-weight historical projection, not to the complete Protocol A 48-candidate selector.

Model	Hard(arith.) − arith.	Hard(geom.) − arith.	Hard(geom.) − geom.	Geom. repair harms
TabICLv2	+0.01831	+0.01800	+0.01836	9/10
TabPFN-3	+0.01178	+0.01126	+0.01192	9/10

6.16. Targeted clean replication and provenance checks

A fresh-runtime check reproduced all six selected configurations at the level of the archived prediction arrays (Table 14). Both joint NLL and Brier values were unchanged at the displayed precision, and the maximum absolute elementwise probability difference was zero. These are six checks of previously evaluated configurations, not six additional independent samples. The recorded environment was Python 3.13.15, PyTorch 2.11.0+cu130, TabICL 2.2.0, TabPFN 9.2.0 and an NVIDIA Tesla T4. The newly configured TabICLv2 checkpoint-path SHA-256 was `bdc7dbd5...b3d4ed0`; the TabPFN-3 V3 checkpoint-path SHA-256 was `d0d865d5...18f3988`. Full fingerprints are retained in the run records. These observations confirm the identity of the configured checkpoint files in the new run, not the bytes actually loaded in all earlier experiments.

In the first online provenance check, downloaded raw-file MD5 values matched the OpenML metadata for ten of eleven source records; the MIC metadata request (OpenML ID 46980) returned HTTP 412. A subsequent standalone retrieval of the official MIC version 3 record succeeded (file ID 22125397): the downloaded 771,864-byte ARFF was reported to match the published MD5, and the required target attribute names were present. The follow-up ZIP retains the verification JSON, including its downloaded-file SHA-256, but not the raw ARFF bytes; therefore we treat this last match as a documented external verification rather than an independently repeated download in this audit. Together, the archived reports document matching MD5 values for eleven raw-source records. Neither raw-file checksums nor the six selected replication runs establish byte-identical preprocessing or checkpoint provenance for all original GPU executions.

Table 14
Independent six-job clean replication, each with context 256, seed 23 and split index 0. Archived and replicated arithmetic-pooling NLL are identical; maximum absolute difference over saved probability tensors is 0 for every row.

Dataset	Model	Archived NLL	Replicated NLL
Car	TabICLv2	1.319614	1.319614
Car	TabPFN-3	1.322587	1.322587
Nursery	TabICLv2	1.348436	1.348436
Nursery	TabPFN-3	1.375204	1.375204
Diamonds	TabICLv2	2.561353	2.561353
Diamonds	TabPFN-3	2.534167	2.534167

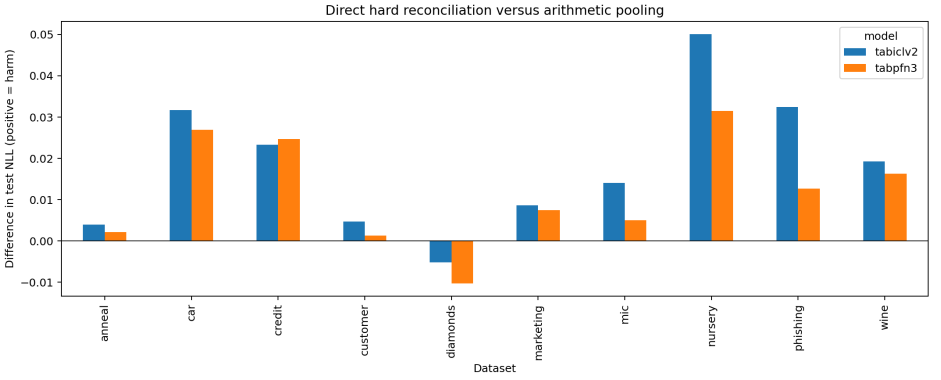
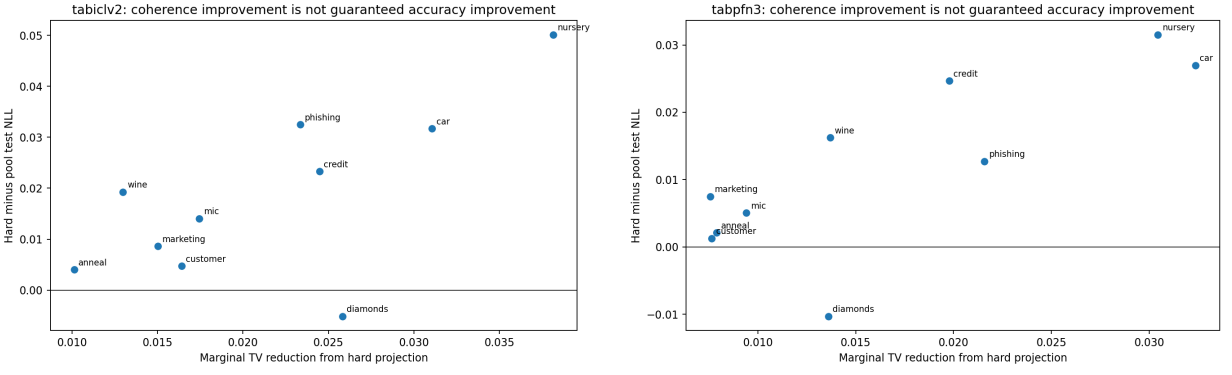


Figure 9: Independent fixed-test audit of hard-reconciliation NLL change by dataset and model at training cap 256. Positive differences indicate degraded predictive log loss after repair. Each dataset value is averaged across four randomized split settings.



(a) TabICLv2 (b) TabPFN-3

Figure 10: Consistency–accuracy diagnostics from the audited 160-cache study. Results are separated by model; reduced disagreement with the model's own marginals does not imply improved held-out log loss.

7. Discussion

Marginal reconciliation solves a well-defined numerical problem. For a positive joint probability table, a projection can adjust its entries until they agree with prescribed row and column marginals. What that calculation cannot determine is whether the prescribed marginals deserve to be trusted. In CoRe-TFM, those constraints come from additional predictions of the same model, not from an observed ground-truth distribution.

The controlled experiments isolate this reliability assumption. When the direct marginal estimates are reliable, their preservation can improve expected log loss. If the conditionals are more reliable instead, the same projection can transmit marginal error into an otherwise useful joint forecast. This change of sign is not a numerical instability of iterative scaling; it follows from which imperfect source is given authority in the objective.

The fixed-test study makes the predictive consequence observable without access to the true joint distribution. Arithmetic-reference hard projection reduced disagreement with the queried marginals in every primary comparison, as its construction is intended to do. It also increased held-out NLL in 67 of 80 comparisons. A second projection using the geometric reference produced a similar dataset-level pattern. Neither result says that marginal consistency is undesirable. Rather, they show that consistency with a model's own imperfect predictions is not a sufficient test of predictive improvement.

There are limits to the pattern. On Diamonds, hard repair improved mean NLL for both models, and the split-level results were heterogeneous. TabICLv2's favorable average, in particular, was driven by one larger improvement against three small deteriorations. A decision rule that uses reconciliation selectively would need to distinguish such cases before seeing the test labels. The guarded rule examined here never activated. Its fallback avoided imposing a repair in those evaluations, but there is no evidence that it learned to recognize beneficial repairs.

The simplest validation-guided policy—choosing from the two raw factorizations and their pools—had the lowest mean NLL in the ten-dataset comparison for both TFMs. Its advantage over arithmetic pooling was small, however, and the adjusted exploratory comparisons did not establish it as a general improvement. The more elaborate candidate family offered no consistent additional benefit. This is unsurprising when validation labels are scarce: a larger search space may create more apparent winners without improving the choices that survive testing (Cawley and Talbot, 2010). The earlier five-seed experiment reaches the same caution through a different measure, separating oracle opportunity from selection regret.

The two real-data protocols answer different questions. The historical budget study changes training context and available selection evidence together. Its sign transition therefore cannot be attributed to either change alone. In the matched fixed-test design, context size changes while the validation and test identifiers remain fixed. Arithmetic pooling improves in 39 of 40 paired split settings for each model when the context grows from 128 to 256 examples. That result concerns additional training information, not reconciliation. The protocols also use different model versions and candidate sets; we do not pool their estimates to manufacture a larger sample.

These findings point toward a practical evaluation standard for probabilistic post-processing. Structural diagnostics are useful for identifying conflicting views, but they do not determine which forecast should replace them. Any repair rule must earn its use through proper scores on genuinely held-out labels, adequate class support, and stability across independent datasets. The evidence here supports that evaluation principle more strongly than it supports any particular automatic repair gate.

Reproducibility should be judged with similar care. The archived probability arrays, scores and split IDs have been audited, and six selected configurations reproduced their stored predictions exactly in a fresh GPU environment. The new runs also recorded checkpoint-file fingerprints. These checks support the numerical findings but do not authenticate the loaded model weights of every historical job. The later MIC raw-source report accounts for the previously missing OpenML checksum verification, yet does not, by itself, establish that all historical feature transformations can be reconstructed byte for byte.

8. Limitations

The formulation is pairwise and categorical. Global compatibility among three or more targets can impose constraints that are not visible in pairwise reconciliation, and mixed continuous targets require different numerical treatment. Extending CoRe-TFM to graphical multi-target reconciliation is future work.

The primary real-model claim covers only two released TFMs. CatBoost is intentionally treated as a boundary baseline rather than a third TFM. TabFM support exists in the codebase, but the frozen protocol excluded it before

comparative outcome inspection for recorded operational-compatibility and runtime reasons. Because the archive contains neither completed TabFM folds nor a controlled TabFM timing study, the paper makes no claim about TabFM accuracy or intrinsic computational efficiency. Broad statements about all TFMs would therefore be unsupported, and replication with additional released TFMs remains a priority.

The datasets are small to medium tabular benchmarks and the protocol is explicitly bounded. The 1024-row cap is not attainable for every dataset, causing the effective training/validation counts to vary at the largest budget. The context experiment also couples upstream training context with validation size, preventing causal separation of the two effects.

The conditioning procedure fits separate estimators for each of four target views and enumerates hypothetical category values at inference. Conditioning on enumerated outcome categories can create feature configurations that are uncommon within the training context, and conditional probabilities from separate fits need not be mutually compatible. The training-pool class-coverage frontloading also creates deliberately non-random nested training subsets, which can favor coverage over representative sampling. These features belong to the experimental protocol rather than being neutral implementation details. It consequently incurs extra fitting and prediction cost relative to a single target classifier. Cached wall-clock values include possible checkpoint loading and download, so they are not valid steady-state throughput comparisons. No full latency, memory, energy or production-cost benchmark is claimed here.

Ten datasets, rather than the much larger number of model–split cells, define the independent units for the main comparisons. The three-seed sensitivity runs above 256 examples are descriptive, and the exploratory policy comparisons were not preregistered as confirmatory tests.

Rare classes remain challenging despite the exclusion sensitivity. Post-hoc support filtering cannot replace a purpose-built rare-event benchmark. Likewise, ECE is binning dependent and is not used as the main inferential endpoint. The controlled reliability mixture is a designed stress test, not an estimate of how frequently different reliability regimes occur in deployed applications.

The oracle uses test outcomes and is included only to illustrate the best score available within the candidate family; it is not a deployable policy. The evaluated guarded rule always selected its fallback, so these data provide no evidence that it identifies cases where applying a repair would help.

Limits of the matched fixed-test evaluation. The independent numerical audit checks cached probabilities, sample IDs and proper scores, not an end-to-end raw-data-to-model rerun of all 160 configurations. The six fresh replications authenticate their newly configured checkpoint-file fingerprints; they do not establish which weights every original job loaded. Eleven raw-source checksums are documented, including MIC, but the historical preprocessing chain remains incompletely verified, particularly for Diamonds. The validation reserve of 205 labels is separate from the 128- or 256-example training context and does not represent a fixed-total-label learning budget. Inference is therefore limited to ten dataset-level units. After exploratory Holm adjustment, hard repair does not meet the conventional 0.05 threshold. The geometric-reference projection was compared after the first results were available and should also be treated as exploratory. No steady-state latency or deployment study is available.

9. Reproducibility

The implementation is organized into model adapters, probability extraction, proper-score metrics, reconciliation modules, benchmark auditing, robustness runners, and reliability-analysis utilities. Unit tests cover reconciliation, soft scaling, selective policies, metrics, synthetic truth, benchmark auditing, robustness execution, and research extensions.

The five-seed benchmark records 1,200 method rows over 150 model–fold tasks, environment versions, fold manifests and checksums. Such provenance is in the spirit of documented machine-learning benchmarking infrastructures such as OpenML (Vanschoren et al., 2014). Supplementary Material S1 extends this evidence to five 256-row seeds and twelve additional context-budget runs (three seeds at each of four other caps), plus selection-family ablations, validation-fraction studies, reliability summaries, rare-class analyses and CPU-only controls. Across all 17 runs it records 20,400 method rows. Dataset–model–fold–seed cells appear eight times, once for each summarized method; those rows are not independent observations for significance tests.

The robustness preflight records Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, TabICL 2.1.1, TabPFN 8.4.0, CatBoost 1.2.10, and an NVIDIA Tesla T4 environment for the archived execution. The benchmark source commit is pinned independently of later manuscript and analysis revisions.

The code is available at `https://github.com/bnssaanirudh/core-tfm`. Robustness figures are drawn from the archived S1 experiment records, rather than from a repository status summary. S1 includes the complete 17-run

aggregate, individual fold-result tables, run-completion records, ablations and environment details. Its 169 listed file SHA-256 values and sizes match the corresponding archived files, and recomputation of the ten dataset effects at the 256-row cap reproduces +0.0008510181 and $p = 0.10546875$. This checks archive integrity and the arithmetic of stored results; it does not constitute a fresh rerun of model inference. The historical S1 evidence is a separate study and is not part of the fixed-test S2 archive cited below. Its independent availability and release identifier must be documented separately.

Matched fixed-test evidence and independent audit. The matched-context study provides two complementary archived objects: a combined evidence ZIP retaining 160 original per-configuration NPZ/JSON prediction caches, 40 frozen split descriptions and associated summaries, and an independent numerical-audit ZIP containing directly reconstructed policy scores, paired intervention effects, dataset-blocked uncertainty, figures, integrity records and environment/provenance-gap reports. The combined manifest lists 414 verified source entries, while the audit bundle lists 23 verified output artifacts. The saved-output audit reproduces 1,920 method-level evaluation rows. Its recorded environment includes TabICL 2.2.0, TabPFN package 9.2.0, PyTorch 2.11.0+cu130 and Tesla T4 hardware; Protocol A used different package versions. Hash and score agreement establish artifact integrity and arithmetic reproducibility, *not* that the exact original weights or raw dataset bytes were independently identified. The fixed-test S2 evidence archive is identified by its Zenodo record (`https://doi.org/10.5281/zenodo.23507196`). This DOI is specific to S2; it does not establish access to the historical S1 artifacts. Neither evidence stream should redistribute credentials or restricted model weights. The numerical audit reads saved arrays without unpickling untrusted processed datasets.

Clean replication and raw-data verification. In six targeted GPU replications, we re-evaluated Car, Nursery and Diamonds under the same saved sample IDs and 256-row contexts, with both primary models. The six probability arrays were identical to the stored historical arrays, and the reconstruction of test NLL and Brier produced zero differences at numerical precision. The replication report includes configured checkpoint-file SHA-256 values, package versions, CUDA hardware and per-job status. These replication records are not backfilled into the older model metadata. An initial online OpenML provenance check matched reported MD5 checksums for ten of eleven raw source records. The remaining MIC record (OpenML 46980) was subsequently retrieved in a standalone check after an earlier HTTP 412 response; its official version 3 metadata lists file ID 22125397, and the separate verification report records a downloaded-file MD5 match and SHA-256 fingerprint. The report is a provenance record, not the raw dataset itself; the older experiment's exact preprocessing identity is not inferred from it. No frozen analysis result was altered. Six clean replications have not changed the original $n = 160$ experimental count.

10. Conclusion

Four probability queries made to the same tabular foundation model need not describe one joint forecast. Reconciliation can make them compatible, but that mathematical improvement is not necessarily a predictive one. The controlled experiments explain the dependence on marginal reliability; the real-model studies show its consequences under held-out scoring.

In the ten-dataset fixed-test benchmark, hard marginal projection reduced structural discrepancy in all 80 primary comparisons and increased NLL in 67. Mean increases were 0.01831 NLL for TabICLv2 and 0.01178 for TabPFN-3. The finding persisted when the projection reference changed, but it was not universal: Diamonds showed favorable average repair effects. The exploratory tests, especially after multiplicity adjustment, do not warrant a blanket claim that hard repair always harms prediction. Neither did a larger selection family consistently outperform simple raw-view-or-pool selection, and the guarded repair policy did not activate.

Coherence and forecasting quality should therefore be evaluated separately. For systems that assemble overlapping probabilistic predictions, satisfying an identity is evidence of internal agreement, not of correctness. Future work should test gates that estimate when particular marginals are trustworthy, extend the analysis beyond two categorical outcomes, and evaluate those decisions under genuinely external distribution shifts.

Table 15
Rare-class exclusion sensitivity at the 256-row cap. Positive effects mean Selective CoRe is worse than arithmetic pooling.

Minimum support	Datasets kept	Mean effect	Wins	Wilcoxon p
1	10	+0.0009	2	0.105
2	9	+0.0008	2	0.164
5	6	+0.0010	1	0.438
10	5	+0.0007	1	0.625

A. Rare-class sensitivity analysis

B. Dataset-level comparisons

The following paired effects are computed from the archived 256-row real-model results after averaging over folds, both primary TFMs and the five seeds within each dataset. The test uses these ten dataset effects, not the 500 underlying TFM model–fold–seed cells.

C. Replication and source-data provenance

The clean replication was independent of policy selection and reproduced previously evaluated configurations. The software environment was Python 3.13.15, PyTorch 2.11.0+cu130, TabICL 2.2.0, TabPFN 9.2.0, NumPy 2.1.3, pandas 2.2.3, SciPy 1.16.3 and scikit-learn 1.6.1 on an NVIDIA Tesla T4. The replication model-file SHA-256 fingerprints were:

```
TabICLv2: bdc7dbd5e4ff21f8f0456fcf90c6b7cdf72dbea960f2d05b19bec19f9b3d4ed0
TabPFN-3: d0d865d54dfbc524f5703104be90620182dca7e5fb2c16de72e9959ea18f3988
```

For TabICLv2, the configured checkpoint repository revision was `4dcd344ece2c00be9e831fdd35bed57b5ad83e19`. For TabPFN-3, the explicit version was V3. These identifiers describe the fresh run only. The archived historical cache sidecars continue to record an unverified checkpoint-weight SHA-256.

The initial online provenance archive retains SHA-256 values and metadata for its ten matched source records. The MIC verification JSON records the official and downloaded MD5 values `4af33fbb77eee0c5600aaa424efba5bc`, raw-file SHA-256 `402ab2d964ed2c23da168e1a6d264f76926fc9365191f492b32b7bbe3c7f08fe`, file ID 22125397, and an ARFF length of 771,864 bytes. The downloaded ARFF was not included in the small verification ZIP; consequently its byte-level digest has not been independently recomputed during manuscript preparation. The six clean-run events remain in the replication record. We do not infer that every historical processed input or checkpoint was byte-identical to the newly verified sources.

Table 16
Five-seed dataset-level Selective-CoRe minus arithmetic joint NLL at the 256-row training cap. Negative values favor Selective CoRe. Values are shown with additional precision to facilitate an independent reconstruction of the paired statistic.

Dataset	Mean paired difference
Car	−0.007898733
Nursery	−0.000549423
Wine	+0.000845628
Customer	+0.000921804
Diamonds	+0.001012085
Marketing	+0.001264137
Credit	+0.001606358
MIC	+0.001727603
Anneal	+0.002186410
Phishing	+0.007394310
Dataset mean	+0.000851018

Table 17
Raw OpenML source provenance outcomes in the independent online check. Wine draws on two source IDs, giving eleven records for ten paired datasets.

Paired dataset	OpenML ID(s)	Raw-record verification
Anneal	46906	MD5 matched
Car	40975	MD5 matched
Credit	46918	MD5 matched
Customer	46938	MD5 matched
Diamonds	42225	MD5 matched
Marketing	46940	MD5 matched
MIC	46980 (v3)	MD5 matched in follow-up check
Nursery	26	MD5 matched
Phishing	46963	MD5 matched
Wine	40691, 40498	Both MD5 values matched

Data and code availability

The implementation is available at `https://github.com/bnssaanirudh/core-tfm`. The machine-readable evidence for the fixed-test study (S2), including saved predictions for 160 model configurations, frozen evaluation splits, audit tables and verification scripts, is identified by Zenodo DOI 10.5281/zenodo.23507196. The five-seed historical study (S1) is a separate evidence stream and is not included in that deposit; its supporting artifacts are described in the reproducibility section. Additional numerical tables and methodological details are included in this article. Source OpenML datasets and model weights are not redistributed in the S2 archive. The six-run replication and raw-source checksum reports provide supplementary provenance, but do not establish byte-identical historical preprocessing for all datasets.

CRediT authorship contribution statement

Badampudi Agasthya Anirudh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Harshita Bogineni:** Methodology, Investigation, Validation, Data curation, Formal analysis, Writing – review & editing. **Sri Preethaa K R:** Supervision, Methodology, Project administration, Validation, Writing – review & editing.

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